



Influence of the time-task constraint on ocular metrics of semi-elite soccer players

Vicente Luis-del Campo ^a, Jesús Morenas Martín ^a, Juan Luis León Llamas ^a, Juan Francisco Ortega Morán ^b,
Jesús Díaz-García ^c and Tomás García-Calvo ^c

^aFacultad de Ciencias del Deporte, Laboratorio de Aprendizaje y Control Motor. Universidad de Extremadura. Avda. de la Universidad, Cáceres (Cáceres), Spain; ^bUnidad de Bioingeniería y Tecnologías Sanitarias, Centro de Cirugía de Mínima Invasión Jesús Usón, Cáceres, Spain; ^cGrupo de análisis comportamental de la actividad física y el deporte. Facultad de Ciencias del Deporte Universidad de Extremadura, Avda. de la Universidad, Cáceres (Cáceres), Spain

ABSTRACT

This study novelty aimed to investigate the influence of manipulating the available time to perform the training tasks on soccer players' ocular metrics, following training. Specifically, pupillary response (pupil diameter) and saccadic features (latency, accuracy, velocity, and number) were measured with a portable eye tracker following training to reflect the mental load accumulated by players during the training sessions. Nine semi-elite soccer players performed two training sessions, based on large-sided games, on an artificial grass field. These two sessions were composed of the same tasks but varying the required time to complete the task goals (*Session 1*: No time limitations to perform the tasks; *Session 2*: Limited time to perform the tasks). The participants performed, before (pre-test) and after (post-test) each training session, a prosaccade task in a room near the playing field. Findings revealed a differentiated effect of the available time to complete the training tasks on ocular metrics because significant differences were found in all variables after training ($p < .001$ for pupil diameter; $p < .01$ for saccade accuracy and number of saccades; $p < .05$ for saccade velocity and latency). Ocular metrics could be a promising tool to evaluate mental load following practice because they were sensitive to the time-task constraint, providing researchers a valuable information for a better planning of the mental workload when designed training tasks.

ARTICLE HISTORY

Accepted 18 January 2023

KEYWORDS

Informational constraint; cognitive load; eye tracking; soccer

Introduction

The use of tasks constraints on soccer performance

The use of training strategies in team sports promotes the fulfillment of the performance outcomes during the competition, and decreases the probability of injuries, diseases, and/or low adaptation to the training (Fox et al. 2018). In many tactical situations, the athletes need to make fast and complex decisions under time pressure and high physical demands (Zurutuza et al. 2017). For example, the interaction between time constraints and information available in the sport environment elicits situations with high cognitive demands that could impair athletes' performance (Gigerenzer and Todd 1999).

Coaches are promoted to afford learning experiences, designing representative training tasks that support the development of athletes' skills (i.e., the design of game-like training tasks; see Davids et al. 2008). In doing so, coaches are required to manage properly different types of constraints to enhance the preparation of their athletes (Van Cutsem and Marcora 2021). Therefore, the coaches should guide the behaviour of their athletes towards performance solutions (Woods et al. 2020), by manipulating the constraints of the training tasks (Renshaw and Chow 2019).

Task constraints are relevant for learning and performance in sport because they can be most easily manipulated by coaches

during practice (Araújo et al. 2004), promoting functional couplings of perception and actions between players and the game environment (Araújo et al. 2009). The use of a temporal limitation performing the training tasks is considered as a non-specific task constraint because no information is prescribed to the players about the action solutions to be performed in achieving the task goals (Balagué et al. 2019). Therefore, the time constraint is related to the initial instructions and rules provides by the coaches during training (Newell 1986).

Surprisingly, studies in soccer manipulating the effect of time constraint on players' performance is scarce although they usually perform their actions under a high time pressure. For an exception, Ponce-Bordón et al. (2022) found that less available time to perform the training tasks significantly increased the internal load perceived by semi-professional players through the Rating of Perceived Exertion (using the CR-10 scale: from 0-rest to 10-maximal) and the adaptation of the NASA – Task Load Index Questionnaire to quantify the subjective feelings of mental load (with responses ranged from 0-no effort perceived to 100-maximum effort perceived). Thus, these increases in values of cognitive load were also accompanied by a decrease of their external physical load and aerobic effort values, compared to the same training but without limitations of time to complete the goals of the tasks.

CONTACT Vicente Luis-del Campo  viluca@unex.es  Facultad de Ciencias del Deporte, Laboratorio de Aprendizaje y Control Motor. Universidad de Extremadura, Avda. de la Universidad s/n 10003, Cáceres, Spain

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Ocular metrics as indicators of mental load

Eye movements are considered reliable indicators of the workload regarding the visual demand of the task (Tao et al. 2019). For example, saccades and pupil size variation are some ocular metrics related to cognitive workload (Mahanama et al. 2022). Pupillary variations in psychological research can be used as an indicator of central nervous system functioning in response to task stimuli (Granholm and Steinhauer 2004). Pupil size could be used to estimate visual attention to the periphery (Brocher et al. 2018; Klatt et al. 2021). Indeed, pupil size is a measure of mental effort because it would reflect within-task, between-task, and between-individual variations in processing demands (Beatty 1982). Similarly, pupil diameter is a relevant indicator of the task difficulty and mental effort (Eckstein et al. 2017). For example, pupillometry was used to address mental workload in soccer players, resulting that a less cognitive effort (i.e., lower pupil diameter) was associated with higher values of tactical knowledge (Cardoso et al. 2019) and better tactical behaviour efficiency (Cardoso et al. 2021).

Saccadic eye movements are associated with different cognitive processes (e.g., attention, processing speed, and inhibitory control; see Hutton 2008). Specifically, saccade-related measures have been found informative in relation to cognitive workload (Keskin et al. 2020). For example, latency of saccades provides insights about cognitive control capacity (Luna and Velanova 2011), or an increase of saccade velocity would indicate a high task difficulty (Di Stasi et al. 2010). Specifically, prosaccadic eye movements in athletes are found to be faster than non-athletes (Lenoir et al. 2000; Di Russo et al. 2003). Yilmaz and Polat (2018) reported a differentiated prosaccadic velocity of athletes depending on what type of sport they accumulated experience. Zwierko et al. (2019) found that the soccer players increased the saccade duration and decreased the saccade velocity, after performing an intense physical effort, during a free-viewing visual search task. Recently, Zhou (2021) showed that skilled female soccer players increased the spatial accuracy of their saccades, with also shorter latencies, compared to the low skilled counterparts.

Based on these findings, it seems that some specific eye movements (saccades) and pupillary response (pupil size) could reveal valuable information related to cognitive processes (e.g., mental workload or visual attention; see Skaramagkas et al. 2021). These ocular metrics would have several advantages for the recognition of the cognitive load compared to other subjective measures (e.g., the NASA-Task Load Index, the Subjective Workload Assessment Technique, or the Rating Scale of Mental Effort; see Charles and Nixon 2019). These physiological measures are also objective evaluations of the mental load that can be standardized and used for comparison between studies, requiring small sample sizes and providing more accurate reports of this cognitive process. According to Charles and Nixon (2019), variables related to ocular metrics are more frequently used in studies of different professional domains due to the development of wearable technology (e.g., portable eye trackers).

The use of eye-tracking devices for the analysis of athletes' gaze behaviour has considerably increased over the last years (Kredel et al. 2017; Hüttermann et al. 2018). Specifically, the

analysis of visual patterns in soccer players has primarily relied upon eye-tracking technology (McGuckian et al. 2018). Thus, the manipulation of task constraints during the training sessions (e.g., game rules, type of equipment used, size and/or marks of the playing areas, nets, and goals, the number of players involved in the practice task) is a common strategy used by the coaches to promote the ongoing search for movement solutions by players to achieve the specific task goals (Renshaw and Chow 2019). However, there are not currently evidences of the time pressure impact on ocular metrics to evaluate mental load of soccer players using portable eye trackers.

Altogether, the aim of this study was twofold: 1) to novelty address the influence of using a time-task constraint on the ocular metrics of semi-elite soccer players following practice and 2) to compare these metrics when they performed a session with timed training tasks (i.e., with a time limitation to solve the goal) and another session with untimed training tasks (i.e., without time limitations to solve the goal). We assume that players' ocular metrics tested following practice would be affected differently by changes in the available time to complete the training tasks. More precisely, we predict that the training session with a time limitation solving the task goals would increase mental load of players, reflected by larger pupil diameters, longer saccadic latencies, higher saccade velocities, lower saccade accuracy, and number of saccades, compared to the session without this time-task constraint.

Methods

Participants

A total of nine experienced soccer players took part in the study ($M_{\text{age}} = 20.04$, $SD = 3.47$). They all accumulated experience playing soccer for more than 0 years prior to the study. They trained four sessions per week (1 hour and 30 min per session). According to the classification system proposed by Swann et al. (2015), the participants were semi-elite players because they competed at national level, but not at the highest level of the soccer leagues existing in their country. They were recruited from the first soccer team of a semiprofessional soccer club. This team was formed by 22 soccer players. The participants initially guaranteed their temporal availability to perform a visual test, before and after each training session. In this line, the research team required them to be on the soccer field 30 min before the warm-up of the training session to perform the visual test (pre-test) and stay on it after training (while performing some stretching exercises) until they were asked to complete again the visual test (post-test).

Participants had normal vision with no reported color blindness or binocular problems, injuries or medical conditions that would exclude participation. They voluntarily took part in the study and gave written informed consent to participate in the study, which was undergo to the guidelines of the University's Ethics Committee and the Declaration of Helsinki. Specifically, this study received approval from the Bioethics and Biosecurity Committee (approval number: 93/2020). Participants received general information about the study objective but were naïve to the specific research contexts and hypotheses.

Instrumental

We used an SMI eye tracking glasses 2w (SensoMotoric Instruments, SMI) to record pupil diameter and prosaccadic eye movements of soccer players sampled at 60 Hz. Subsequently, these ocular metrics were extracted throughout the SMI BeGaze software. The portable eye-tracking device was also used to control that the players' gaze was initially fixated at the central cross of the computer screen before the appearance of the target stimulus (a black dot). Specifically, we measured the pupil diameter (in micrometers during the trials, and averaging both eyes), saccadic latency (i.e., the delay of time between the appearance of the visual target on the screen and the onset of the saccade towards this stimulus), average saccadic velocity (i.e., the average saccade speed in degrees per second), saccade accuracy (i.e., the ratio of correct go and no-go decisions across all trials for each player) and total number of saccades made during the visual test. Based on past studies, we used these ocular metrics for their relations with mental efforts (Tao et al. 2019; Mahanama et al. 2022).

We also used the Polar Team Pro to measure the mean heart rate (mean HR) of the soccer players in both training sessions (unit: beats per minute or bpm). Specifically, this instrument allowed to quantify objectively the physical load performed by the participants in each training.

Design and procedure

The study involved a cross-sectional research design with participants completing two training sessions on an official soccer field with an artificial turf surface, separated one week apart from the other. Both trainings were based on large-sided games to simulate full game or game phases in team sports, changes in tactical behaviors of players but also variations in their external and internal load (Hill-Haas et al. 2010). Each training session was formed by: i. 2 minutes long 8 × 8 possession games (40 × 40 m.) in which the goal was achieved after performing 10 consecutive passes by team (the game continued after players completed the 10 passes or if there was a turnover with a new count); ii. 2 minutes long 8 × 8 possession game (60 × 40 m.) in which the goal was achieved after performing 10 consecutive passes by team (the game continued after players completed the 10 passes or if there was a turnover with a new count); iii. 6 minutes long matches with goalkeepers and traditional goals (70 × 50 m.); iv. 6 minutes long matches with goalkeepers and traditional goals (100 × 64 m.). Before training, they performed a familiar soccer warm-up of 0 minutes. The resting time between tasks was 5 min.

We varied task conditions between sessions by manipulating the available time to complete the goal tasks. Specifically, players had not a limited time to achieve the purpose of the tasks in the session without time limitation; meanwhile, in the session with time limitation: the value of the goal was double from minutes 3:00 to 6:00 and from 9:00 to 12:00 during (i.) the first possession game; from minutes 3:00 to 6:00 and from 9:00 to 12:00, goal was obtained after six consecutive passes only during (ii.) the second possession game; the value of the goal was double from minutes 3:00 to 6:00 and from 9:00 to 12:00 during (iii.) the first match game; the value of the goal was

double in the case of obtain a goal performing a maximum number of five passes after the ball was recovered during (iv.) the second match game. These modifications were designed to promote changes in the score of the task, less time to revert adverse scores and increases in the pacing of the game.

Regarding this study, we analyzed pro-saccades (i.e., reflexive saccades triggered exogenously by the occurrence of a stimulus in the visual periphery; Guidetti et al. 2019). For this purpose, we used a prosaccade task to examine the mental load of participants during two training sessions throughout the analysis of the pupil diameter and prosaccadic features, as Zhou et al. (2021) previously made with young soccer players. This visual test was performed at a closed room nearby to the soccer field, with similar experimental conditions of temperature, sound, and brightness of previous studies (Cardoso et al. 2019, 2021).

Participants were instructed to fixate at the black cross located at the center of a computer screen as long as the target appeared peripherally. They were seated at 60 cm distance from the screen with their head supported by a chin rest. When the player perceived a black dot at one of the four corners of the computer, they had to initiate a saccade toward the target as quickly and accurately as possible once it appeared (i.e., the prosaccade task under the 'go' condition). Moreover, when the dot was white, the players had to keep their gaze at the central cross of the screen (i.e., the antisaccade task or the 'no-go' condition). The test consisted of a random sequence of 16 trials, with a proportional number of targets to each corner (4 trials on each of the 4 corners), and stimulus color (8 black circles for the prosaccade task and 8 white circles for the antisaccade task). The order of the trials was different, before and after each training session, with an interval of 3–5 s for resting time between trials (Figure 1).

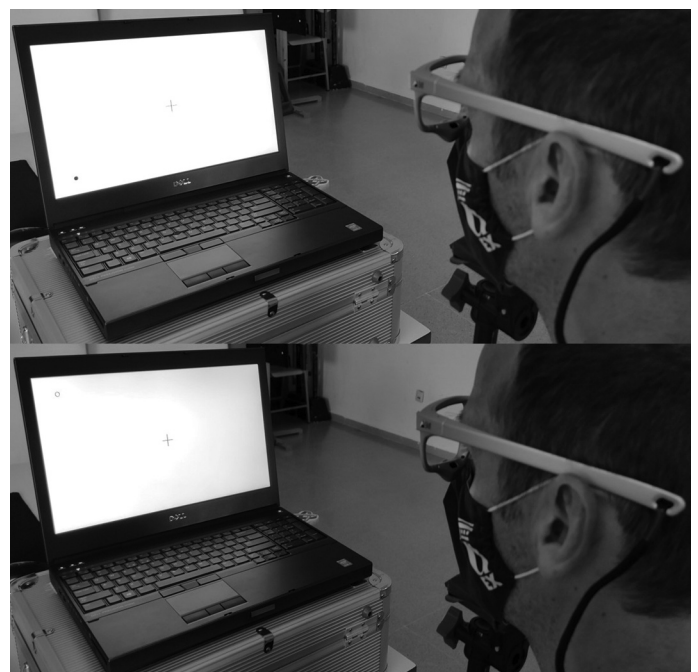


Figure 1. A participant performing the prosaccade task at a closed room nearby to the soccer field.

Participants performed an accurate saccade when performed a prosaccade towards a black dot at one of the four corners of the screen or maintained the gaze shift at the central cross of the screen in case of displaying a white dot. Contrary, this decision outcome was incorrect when the participants performed a prosaccade with a white dot. The duration of the test was 3 min. All soccer players were measured within the 30 min before and after each training session. This procedure ensured just a short rest break for all participants following training, preventing a decrease of the time-task constraint effects on their ocular metrics. We strived to achieve this endeavour because cognitive functions are found to be influenced by the exercise durations and recovery periods (Crush and Loprinzi 2017). For example, the application of a resting time with 30 min of mental recovery may enhance perceptual outcomes and cognitive performance (Loch et al. 2020).

Data analysis

The independent variable was the available time to complete the goal of the tasks during the two training sessions (*Level 1*- The participants had no time limitations in achieving the goals: Untimed session; *Level 2*- The participants had time limitations in achieving the goals: Timed session). The Shapiro–Wilk test revealed that the data of ocular metrics were not normally distributed. Therefore, non-parametric analyses were performed in this study. The U-Mann Whitney test was used to address possible effects of the time-task constraint on ocular metrics between sessions, following training. The Wilcoxon test was used to seek differences in these metrics within each training session, comparing values from pre-training to post-training.

We also measured the physical efforts performed by the soccer players in each training session. We used again the U-Mann-Whitney to compare the values of this variable between sessions. Alpha levels for statistical significance were set at $p < .05$ for all analyses. Statistical analysis was performed using the statistical package SPSS 25.0 (2017 SPSS Inc.).

Results

Table 1 shows significant differences on all variables when compared both sessions, after training. Specifically, the soccer

players displayed higher values of pupil diameter, saccade accuracy, and saccadic latency when performed the training session without time limitation to solve the tasks than the training session including this time-task constraint. On the contrary, the data regarding saccade velocity and number of saccades were higher at the timed session compared to the untimed session.

The players displayed higher pupil diameters and saccade accuracy after training compared to values before training when performed the untimed session. No significant differences were found for these variables when they performed the timed session. The players also decreased their saccade velocities after training during the untimed session. Contrary, they increased the saccade velocity when completed the timed session. Finally, the players displayed slower saccadic latencies and lower number of saccades after training vs. before training, for both training sessions.

The physical load performed by participants in each training session was similar because no significant differences were found in the mean values of HR between sessions ($U = 8247$; $p = .71$). Specifically, the mean HR was 163.8 bpm ± 16.12 (Timed session) and 160.6 bpm ± 14.37 (Untimed session).

Discussion

The aim of the study was to address the effects of a time-task constraint on semi-elite soccer players' ocular metrics when performed training tasks based on large-sided games. This is the first study attempting to analyse the influence of time pressure on pupillary response and saccadic features of players following training because cognitive load has been a human factor rarely used in the sport domain from the analysis of the ocular metrics.

We expected that time pressure would reveal differences on ocular metrics of soccer players depending on whether they performed timed or untimed training tasks because previous studies found that measures regarding pupil size and saccadic eye movements showed responsiveness to mental load (Tao et al. 2019). In line with our initial expectations, the time-task constraint caused a differentiated impact on ocular metrics of players regarding the available time to complete the training tasks because differences were found in all variables when compared the two sessions following practice. Therefore, the

Table 1. Comparison of values (M=mean, SD=Standard Deviation) for ocular metrics displayed by participants when performed the training tasks during the timed and untimed sessions.

Variables		Session without time limitation		Session with time limitation		Between Sessions
		Pre-	Post-	Pre-	Post-	Post-
Pupil diameter	M	2.79	3.02	2.71	2.75	
	SD	.22	.25	.19	.14	
		$\chi^2 = 13.96^{***}$		$\chi^2 = .45$; $p = .65$		$U = 10752^{***}$
Saccade accuracy (%)	M	97	99	98	.96	
	SD	.17	.08	.15	.20	
		$\chi^2 = 2.32^*$		$\chi^2 = 1$; $p = .31$		$U = 53424^{**}$
Saccade velocity (°/s)	M	114.81	112.06	71.39	118.83	
	SD	28.18	39.31	36.59	42.22	
		$\chi^2 = 2.27^*$		$\chi^2 = 11.02^{***}$		$U = 23040^*$
Saccadic latency (ms)	M	308.36	337.50	302.44	312.25	
	SD	81.25	90.75	89.95	88.27	
		$\chi^2 = 4.19^{***}$		$\chi^2 = 3.01^{**}$		$U = 12567^*$
Total number of saccades (count)	M	55	35.40	50.44	41	
	SD	34.36	6.84	19.39	15.48	
		$\chi^2 = 4.85^{***}$		$\chi^2 = 4.35^{***}$		$U = 26112^{**}$

Note: $^{***}p < .001$, $^{**}p < .01$, $^*p < .05$.

ocular metrics used in this study were sensitive to this task constraint during soccer training.

Recently, it has been found that the manipulation of a temporal constraint influenced subjective cognitive responses and individual performance of basketball players. For example, Camacho et al. (2021) concluded that amateur players perceived more mental load when a task constraint was used (e.g., seven seconds possession, three passes maximum per attack phase, or a combination of both) compared to a free game condition during basketball small-sided games. Kostrna (2022) found that undergraduate basketball players of different skill levels achieved a lower field goal percentage of shooting at the timed condition (i.e., to successfully score as many field goals as possible in 30 s) compared to the untimed one (i.e., to attempt the same number of field goals that in the timed condition but without the 30-s time constraint).

Firstly, to highlight that the values of pupil size were closely to 3 mm in both sessions, before and after training, ensuring that the testing of the ocular metrics was warranted within standard light conditions (Wyatt 1995). The post-training differences in pupil size between sessions could reveal that the intensity of the mental load suffered by the soccer players were different during the training tasks. As a result, the soccer players showed a differentiated capacity to look at salient stimuli as result of their different mental effort elicited in both sessions (reflected by significant differences on values of pupil diameter; see Mathôt et al. 2015).

To highlight that the physical efforts performed by the participants were similar in each training session because no differences were found in the mean heart rate when compared both sessions. Therefore, the differences found in this study for ocular metrics were not explained at a physical level. Contrary, we reasoned that the differences in pupil size values and saccadic eye movements were driven by the manipulation of the time pressure suffered by the soccer players in the two training sessions. Specifically, the players showed a smaller pupil diameter at the timed session than the untimed one after training. We initially assumed that this pupillary response would be larger when participants have to solve the training tasks with a time limitation due to an increase of the mental load. However, small pupil sizes have been related to improvements of attentional performance in go/no go tasks (Gilzenrat et al. 2010).

The soccer players also showed a higher saccade velocity and a lower saccadic latency during the timed session compared to the untimed one. We argue that those training tasks with a limited time to complete the goals enhanced the participants' attentional system, resulting in a quick reaction to peripheral stimuli displayed during the visual test (Brocher et al. 2018; Klatt et al. 2021). To our best knowledge, the performance of training tasks with a time pressure drifted towards players' faster saccadic responses to the target stimulus during the visual post-test. We suggest that they were exposed during practice to playing situations with more information processing speed as the time available to extract relevant information from non-foveated locations of the training tasks was lower compared to the same ones but without time constraints. This finding would mean that training tasks with high cognitive demands (i.e., tasks with a time pressure)

enhanced attentional system of players because their ability to quickly shift attention between one visual cue (located at the central cross) to another one (in the visual periphery) was faster compared to the tasks with low cognitive demands (i.e., task with no time limitations).

The saccade velocity only increased when time pressure got higher for the players. Contrary, the session without time limitation decreased this saccade velocity. We reasoned that the higher the saccade velocity, the higher the cognitive load of players because an increase of the velocity to saccade would indicate a higher task difficulty performing the tasks with a time pressure (Di Stasi et al. 2010). Pradhan et al. (2019) stated that high values of saccadic velocities indicated better cognitive performance in young soccer players during a mental workload task. An increase of this saccade velocity has been also found in skilled female adolescent soccer players compared to their less skilled counterparts, as a function of the skill level (Zhou 2021).

However, a negative consequence of a fast saccade to the target appeared peripherally was that the players achieved a lower saccade accuracy at the timed session than the untimed one. Therefore, a lower number of efficient goal-directed saccades were found when the visual test was preceded by training tasks with a time pressure. We reasoned that this task constraint at the timed session imposed a higher mental workload to the soccer players, impairing their visual processing mechanisms for contrast detection and discrimination of the visual target during the visual post-test. In similar vein, a saccade performed too quick could have the bias of not being very accurate due to the signal-dependent noise accompanying the control signal (Harris and Wolpert 1998).

We defend that both sessions increased the mental effort of players after training because they had to repeatedly make decisions in dynamic and changing games and simulated matches, based on specific visual information available in that environment (teammates, opponents, and ball), and perform a set of technical and tactical actions for the accomplishment of the task goals. As a result, the number of saccades decreased after training in both sessions. Pradhan et al. (2019) found a relation between the type of profile of young football players (American football, hockey) and the number of saccades performed during a diagnostic visualization tool. They showed that a slow speed reading numbers (low profile) increased the number of saccades as compared to other moderate and fast profiles of players, both in the pre- and post-sport season. According to these authors, the group of participants with a slow profile had no improvements in saccadic metrics (e.g., number of saccades) between pre- and post-season because of the immaturity of their underlying oculometric systems.

For the untimed session, all ocular metrics significantly changed their values from pre-training to post-training. Similarly, the data of the variables at the timed session varied from pre-training to post-training (except for the pupil diameter and saccade accuracy). Collectively, these results provide evidence that ocular metrics showed responsiveness to the efforts performed by the players during the training tasks, independently of the presence or absence of a time pressure, because differences were found when compared values before and after training, in each training session.

Conclusions

Pupillary response and saccadic features of semi-elite soccer players were affected differently regarding the available time to perform the training tasks. Therefore, the pupil size and measures of saccadic eye movements were influenced by the time-task constraint used by the coaches during training. These findings could reveal a differentiated mental load suffered by the players following practice throughout the analysis of the ocular metrics, as a result of a previous manipulation of the time-task constraint during training.

Limitations and future research

This study has several strengths. First, this research novelty addressed the impact of time constraint on soccer players' ocular metrics during training tasks. Second, these saccadic features and pupillary response offers a challenging methodology to evaluate mental workload in soccer following practice. Third, the use of multiple eye-tracking measures in the same task provided opportunity to address in a better shape the impact of a temporal task constraint on players' cognitive load. The main limitation of the study was the low number of sessions performed by soccer players, and the number of participants performing the test for ocular metrics, masking the possibility to address the generalizability effect to the sampled population of semi-elite soccer players. Another limitation was the lack of a familiarization session with the prosaccade task. However, the two training session period was also the maximum time for attendance and availability that the players agreed to commit for the experiment. Thus, a control group that completed equal physical activity output but did not complete decision-making activities would have needed to address the effects of the physical exertion of the players alone on their cognitive fatigue (measured with ocular metric outcomes) presented in the visual test following training.

Next studies investigating the time pressure in soccer players while training on-field should correlate these ocular metrics with other measures of their tactical and/or technical behaviours (e.g., number of passings made during the tasks or the shootings to the goal, percentage of ball possession), but also with their decision-making performance (e.g., the speed and accuracy of their decisions according to their motor execution), for a better understanding of how player behaviour was influenced by this task constraint manipulation and how this related to cognitive load measured post-training. Thus, further specific variables of the internal (e.g., the heart rate) and external load (e.g., mean and/or maximum speed, number of sprints) should be added for a better control of the physical intensity performed by the participants during the experimental sessions. Similarly, future studies should add variables addressing emotional and/or arousal states of participants because pupil size variations have been revealed as an index of affective processing. To exemplify, pupillary responses could be constrained by the dynamics of team results in competitive matches or by the type of training to perform following the warm-up of sessions (physical training vs tactical training).

Further research is needed to clarify not only the impact of the time pressure but also the direction of this impact on ocular metrics. In future studies, it would be interesting to analyze the fixation-related parameters of eye movements. In doing so, a visual search test to make specific decisions could help in this endeavour (e.g., the tacticUP test; see Machado and da Costa 2020). These complementary eye movement signals could inform about where or at what processing stage the load is imposed on. For example, whether the players perceived an increase of the mental workload, and a decrease of the fixations durations was reported, would this mean a manipulation of the perceptual load? Contrary, whether the players perceived an increase of the mental workload, and an increase of the fixations durations was reported, would this mean a manipulation of the cognitive load (Liu et al. 2022). This information would be useful for a better design of the training tasks to correctly manage the mental efforts of the soccer players during an organized sport season.

Practical applications

Soccer performance is influenced by the ability to maintain visuomotor processing during physical and mental efforts while performing different perceptual and motor skills with a high time pressure. Therefore, coaches should carefully manipulate task constraints during trainings to promote adaptive behaviours of their players in an attempt to achieve the tasks goals. The introduction of time constraints during the training tasks would offer coaches the opportunity to address mental load of players through the use of objective measures as ocular metrics. This information would be useful to guide more accurately their training schedules because they could quantify not only the physical performance of the players but also their cognitive states.

Soccer players must distribute their attention over different objects available at the playing game (e.g., ball, teammates, and counterparts) and process this visual information under time pressure to react quickly to complex and changing situations. Therefore, they need to identify as soon as possible the target stimulus from the task-irrelevant stimuli performing different saccadic eye movements. Within this framework, we encourage soccer coaches to investigate the attentional load of players during training through the analysis of their ocular metrics following practice because these biomarkers could be used as a tool to evaluate, for instance, the allocation of the visuospatial attention. In doing so, the manipulation of the task constraints and their effects on ocular metrics could help coaches to better control the mental load of players during training. Specifically, this information seems of interest to adequately use this constraint in the planification of the mental load during soccer microcycles. Mentally demanding constraints should be avoided close to competitions, searching to avoid possible negative effects of mental fatigue on performance.

The use of portable eye trackers would also provide coaches the opportunity to capture some features of the soccer players' eye movements and pupillary response in real-time. Thus, the easy and quick calibration setup of these instruments affords

them to start recordings of players' gaze within seconds. As a result, the coaches staffs could inspect in a brief time, through the software interfaces of these wearable devices, some measures of the ocular metrics displayed by the players before and after training. The capture of this physiological information from ocular biomarkers would be particularly relevant following training to enhance our understanding of the cognitive demands suffered by the players during training (short time scale) but also to control properly the mental states of the players during a competitive season (long time scale).

We recommend using pupil size and prosaccadic eye movements as reliable objective biomarkers for mental workload. These measures could be also a promising tool for testing specific features of the underlied processes of visual perception and attention to the periphery during training. Additionally, the study of the pupillometry and prosaccades could provide a sensitive indicator of decision outcome because go/no-go decisions would inform about the internal neural activity related to decision-making processes performed by soccer players during this type of decision-tasks.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

The author(s) reported there is no funding associated with the work featured in this article.

ORCID

Vicente Luis-del Campo  <http://orcid.org/0000-0003-3749-3294>
 Jesús Morenas Martín  <http://orcid.org/0000-0002-0814-0636>
 Juan Luis León Llamas  <http://orcid.org/0000-0002-0757-5937>
 Juan Francisco Ortega Morán  <http://orcid.org/0000-0002-0287-5498>
 Jesús Díaz-García  <http://orcid.org/0000-0002-9430-750X>
 Tomás García-Calvo  <http://orcid.org/0000-0002-2550-418X>

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